MCS-218: Data Communication and Computer Networks

Assignment No. MCA_NEW/MCAOL(II)/218/Assign/2026-27 | Maximum Marks: 100 (80 + 20 viva) | Session: July 2026

Name: _____ Enrolment No.: _____ Study Centre: _____

Q1. Network topologies, with advantages and disadvantages (10 marks)

A **network topology** is the physical or logical arrangement of nodes and links in a network.

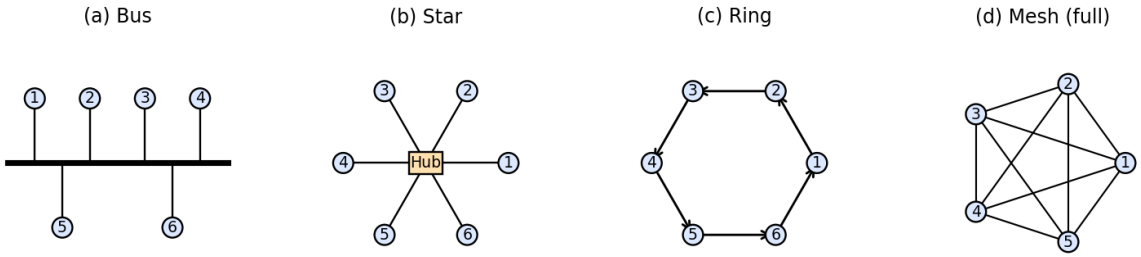


Figure 1: Bus, Star, Ring and Mesh topologies

a. Bus topology

All nodes are attached through drop lines/taps to a single shared backbone cable. A signal sent by one node travels along the bus and is received by every node; only the addressed node accepts it. Terminators at both ends absorb the signal. Access is controlled by CSMA/CD (classic Ethernet).

Advantages	Disadvantages
Easy to install; needs the least cable Cheap, easy to extend by adding a tap Failure of one node does not stop the others	A break in the backbone splits or stops the entire network Performance falls as traffic and nodes increase (collisions) Fault isolation is difficult; limited cable length

b. Star topology

Every node is connected by a point-to-point link to a central hub or switch. All traffic passes through the centre, which repeats (hub) or forwards selectively (switch).

Advantages	Disadvantages
Easy to install, reconfigure and add nodes Failure of one link affects only that node Easy fault detection and management	Central device is a single point of failure More cabling than bus Cost of hub/switch; central device limits size

c. Ring topology

Each node is connected to exactly two neighbours, forming a closed loop. Data circulates in one direction (or both in dual ring) and each node regenerates and forwards the signal. A token is commonly used for access (Token Ring/FDDI).

Advantages	Disadvantages
Orderly access, no collisions (token passing) Equal access for all nodes; good performance under heavy load Each node acts as a repeater, so long distances possible	A single node or link failure can break the ring (unless dual ring) Adding or removing a node disrupts the network Troubleshooting is harder; delay grows with number of nodes

d. Mesh topology

Every device has a dedicated point-to-point link to every other device (full mesh). For n nodes there are n(n-1)/2 links and each node has n-1 ports. Partial mesh connects only some nodes this way.

Advantages	Disadvantages
Very reliable; multiple alternate paths No traffic problems, dedicated links Secure, private links; easy fault identification	Huge amount of cabling and I/O ports Expensive to install and maintain Hard to scale and configure

Q2. Bandwidth of the channel (5 marks)

Bandwidth of a channel is the difference between the highest and lowest frequencies it can pass:

$B = f_{max} - f_{min} = 7000\ Hz - 2000\ Hz = 5000\ Hz = 5\ kHz$

Q3. Differences between modulation techniques (10 marks)

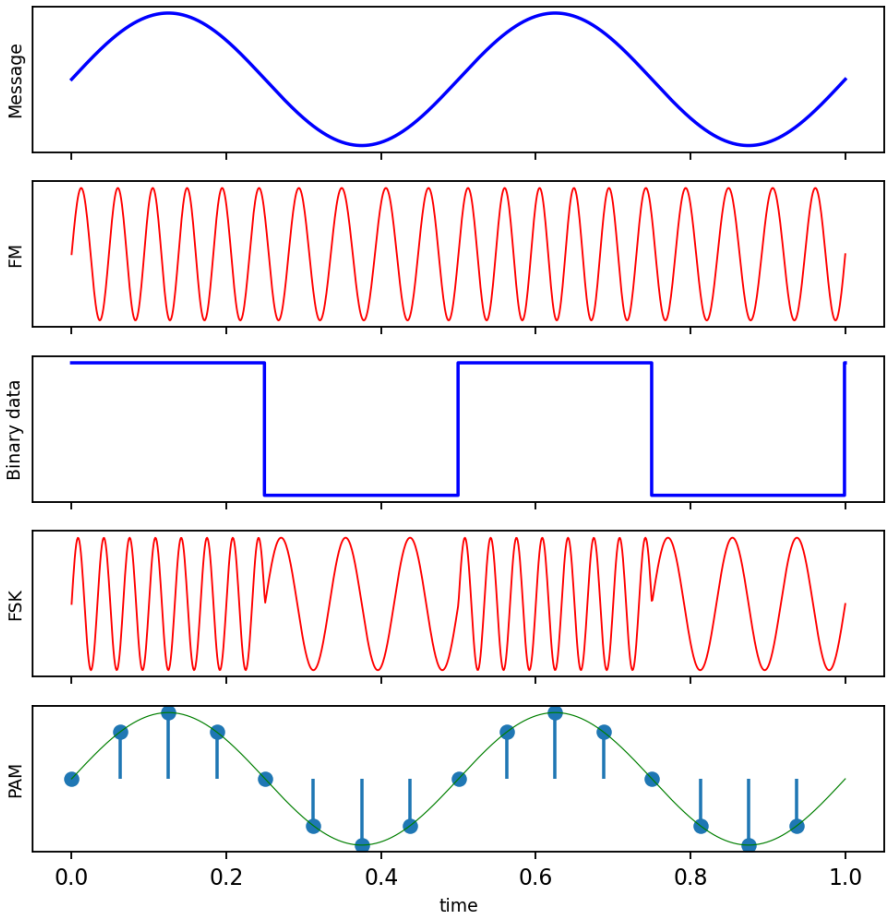


Figure 2: Message signal, FM, binary data, FSK and PAM

Feature	Frequency Modulation (FM)	Pulse Code Modulation (PCM)	Frequency Shift Keying (FSK)	Pulse Amplitude Modulation (PAM)
Input signal	Analog	Analog	Digital (binary)	Analog
What varies	Frequency of a sinusoidal carrier varies with message amplitude; amplitude constant	Analog signal is sampled, quantized and each level encoded as a binary code word	Carrier frequency switches between two (or more) values, one for 0 and one for 1	Amplitude of a train of periodic pulses varies with sampled message value
Carrier	Continuous sine wave	Pulse (digital bit) stream	Continuous sine wave	Pulse train
Output	Analog	Digital	Analog (carrier-modulated)	Analog pulses (discrete time, continuous amplitude)
Noise immunity	Good (noise mostly changes amplitude)	Very good; regenerative repeaters	Good, better than ASK	Poor, like AM
Bandwidth	Wide (Carson's rule 2(delta-f+fm))	Wide (n bits x sampling rate)	Moderate (depends on the two tones)	Same as pulse rate
Typical use	FM radio, TV sound	Digital telephony (64 kbps), CDs	Modems, RFID, low-rate data	First step in PCM, TDM

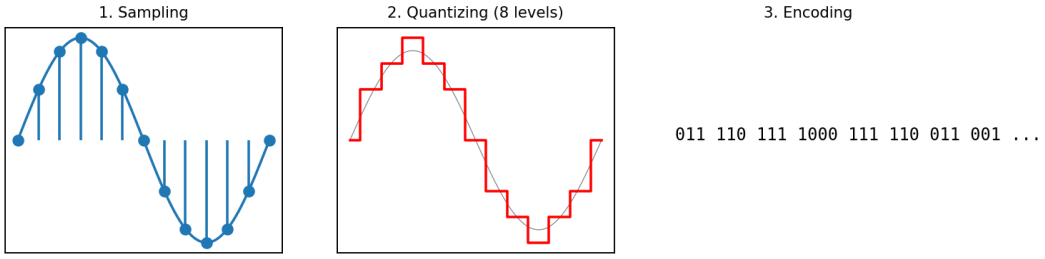


Figure 3: PCM steps - sampling, quantizing, encoding

Q4. Maximum number of amplitude levels (10 marks)

Given: amplitude range = -5 V to +7 V; minimum separation between levels = 0.6 V.

Total voltage span = 7 - (-5) = **12 V**.

Levels are placed at 0.6 V intervals, so the number of intervals = 12 / 0.6 = 20. With 20 gaps we can place 20 + 1 levels (both end points -5 V and +7 V are usable):

Maximum number of levels L = 12/0.6 + 1 = 21

Levels: -5.0, -4.4, -3.8, ... , +6.4, +7.0 V.

Remark: Bits per symbol = $\log_2 21 = 4.39$, so a practical coder uses the nearest power of two, 16 levels (4 bits per symbol), with the extra margin improving noise tolerance. By Nyquist, maximum bit rate = $2B \log_2 L$.

Q5. Switching in communication networks (5 marks)

a. Significance of switching (2 marks)

In a large network it is impractical to connect every pair of devices by a direct link (n(n-1)/2 links). Switching lets many devices share a limited set of links by establishing temporary paths through intermediate nodes. Significance: (i) reduces number of links and cost, (ii) enables efficient sharing of resources, (iii) provides routing of data from source to destination across the network, (iv) supports

scalability and different traffic types (voice, data).

b. Switching techniques (3 marks)

i. Circuit switching. A dedicated end-to-end path is set up before data transfer in three phases: *connection establishment* (signalling reserves a channel on every link), *data transfer* (continuous, fixed delay, no header per data unit), and *connection release*. Resources stay reserved even when idle. Used in the telephone network (PSTN). Implemented with space-division or time-division (TDM) switches. Good for real-time voice; inefficient for bursty data; setup delay and possible call blocking.

ii. Multistage switching. A single N x N crossbar needs N^2 crosspoints, which is too costly for large N. A multistage switch (e.g. Clos network) connects smaller crossbars in stages (input, middle, output). Each input switch has several outputs, one to each middle-stage switch, and each middle switch has one link to every output switch. A call finds a free path through some middle switch. This greatly reduces crosspoints; a Clos network is non-blocking when the number of middle switches m >= 2n - 1 (n = inputs per first-stage switch). Trade-off: possible blocking with fewer middle switches.

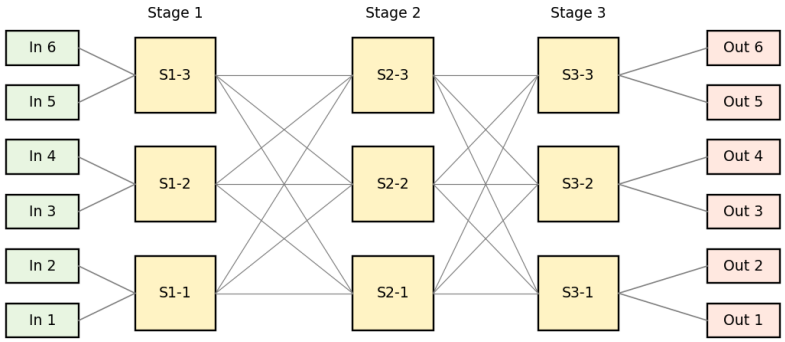


Figure 4: Three-stage (multistage) switch

iii. Packet switching. The message is divided into packets, each with a header (source, destination, sequence number). Packets are stored and forwarded by switches/routers, sharing links by statistical multiplexing. *Datagram* approach: each packet routed independently, may arrive out of order (Internet/IP). *Virtual circuit* approach: a route is set up once and all packets follow it (X.25, Frame Relay, ATM). No dedicated resources, so efficient for bursty data; drawbacks are variable delay, jitter, and possible packet loss/congestion.

	Circuit switching	Packet switching
Path	Dedicated, set up first	Shared, per packet/virtual circuit
Resource use	Reserved even when idle	On demand, efficient
Delay	Setup delay, then constant	Variable (queuing)
Best for	Voice, real-time	Bursty data

Q6. Retransmission strategies: terms (5 marks)

a. Piggybacking (2.5 marks)

In full-duplex communication both stations send data frames. Instead of sending a separate ACK frame, the receiver **attaches (piggybacks) the acknowledgement onto its next outgoing data frame** using the ACK field in the frame header. If no data is ready, it waits a short time, then sends a standalone ACK. This saves bandwidth and reduces the number of frames, improving link utilization.

b. Pipelining (2.5 marks)

In stop-and-wait the sender stays idle until the ACK returns, so utilization on long-delay links is very low. **Pipelining** lets the sender transmit several frames continuously (up to the window size) before waiting for acknowledgements, keeping the channel busy. It is used in Go-Back-N and Selective Repeat ARQ; sequence numbers identify frames and a sliding window controls how many are outstanding. Utilization = W / (1 + 2a), where a = propagation time / frame time.

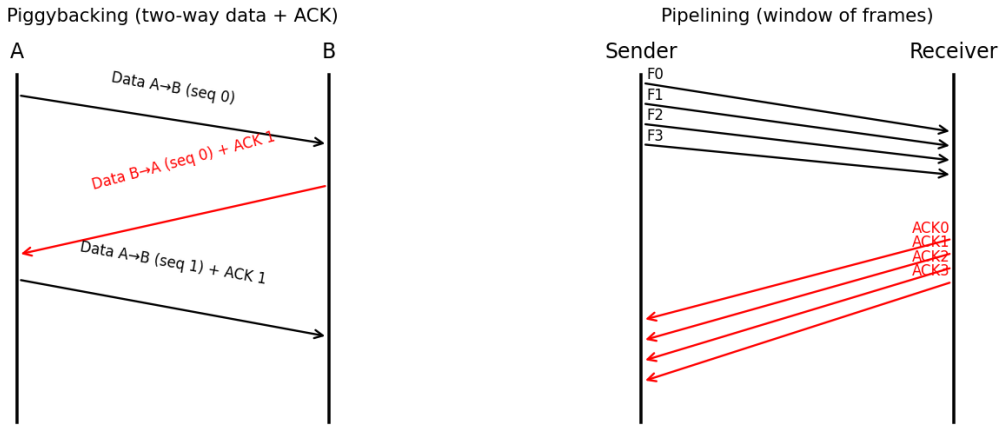


Figure 5: Piggybacking and pipelining

Q7. IEEE 802.11 and Ethernet (10 marks)

a. Fragment burst in 802.11 (5 marks)

Wireless channels have high error rates, so a long frame is likely to be corrupted. 802.11 therefore allows a frame to be split into smaller **fragments**, each acknowledged separately, so only a damaged fragment is re-sent. A **fragment burst** is the sequence of fragments of one MAC frame sent one after another with a single contention for the channel.

- **Step 1:** The sender wins the channel using CSMA/CA (DIFS wait plus backoff); optionally RTS/CTS is exchanged.
- **Step 2:** It sends Fragment 0. The Duration field reserves the channel (NAV) for the next fragment + its ACK.
- **Step 3:** After a SIFS the receiver returns an ACK for Fragment 0. The other stations, seeing the NAV, remain silent.
- **Step 4:** After a SIFS (shorter than DIFS, so no other station can grab the channel) the sender sends Fragment 1, and gets its ACK; this repeats for each fragment.
- **Step 5:** The last fragment has 'More Fragments' = 0; its ACK ends the burst and the channel is released. A lost fragment is retransmitted and the burst continues; if ACK is missing the sender contends again.

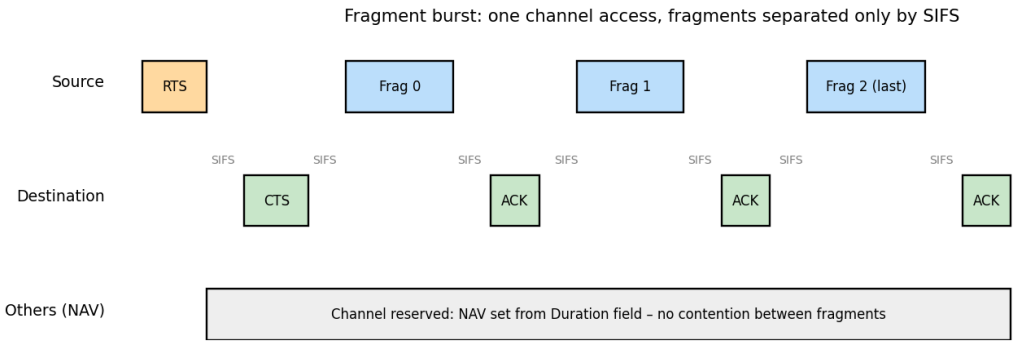


Figure 6: Fragment burst with RTS/CTS, SIFS spacing and NAV

b. Binary exponential backoff in Ethernet (5 marks)

After a collision in CSMA/CD, a station aborts, sends a jam signal and waits a random time before retrying. The algorithm (time slot = 512 bit times = 51.2 µs at 10 Mbps):

- After the *n*-th consecutive collision for a frame, choose an integer K at random from {0, 1, 2, ..., 2^m - 1}, where m = min(n, 10).
- Wait K x 51.2 µs, then sense the channel and retransmit.
- After 1st collision K in {0,1}; after 2nd K in {0..3}; after 3rd K in {0..7}; ... up to {0..1023}.
- After 16 failed attempts the station gives up and reports an error to the upper layer.

The window doubles with every collision, so when many stations are contending the retry times spread out, reducing repeated collisions. Weakness: a station that just succeeded has a small window and may win again (capture effect).

Q8. Routing and flooding (20 marks)

Assumption. Figure 1 as printed shows the flooding time labels (t = 0, 1, 2) on the links and no explicit cost numbers. Hence every link is taken to have **cost 1**. The links are: R1-R2, R1-R3, R2-R4, R2-R5, R3-R4, R3-R6, R4-R5, R4-R6, R5-R7, R6-R7. If your copy shows different costs, repeat the same table with those costs.

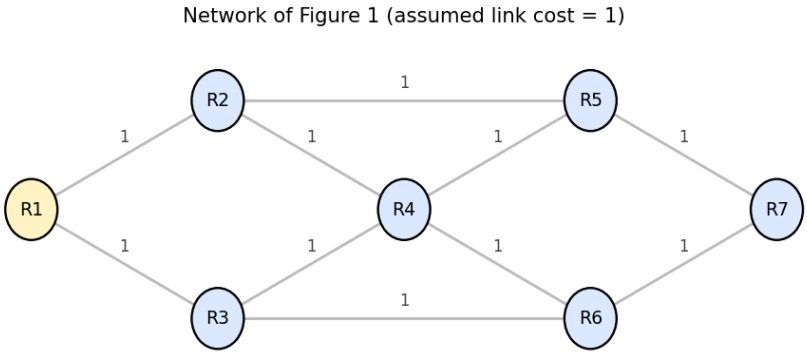


Figure 7: Network topology used (cost 1 per link)

a. Dijkstra's shortest path from R1 to R7 (10 marks)

Notation: each entry is (distance, predecessor). Permanent node is marked *.

Step	Permanent set	R2	R3	R4	R5	R6	R7
0	{R1*}	(1,R1)	(1,R1)	inf	inf	inf	inf
1	+R2	1*	(1,R1)	(2,R2)	(2,R2)	inf	inf
2	+R3		1*	(2,R2)	(2,R2)	(2,R3)	inf
3	+R4			2*	(2,R2)	(2,R3)	inf
4	+R5				2*	(2,R3)	(3,R5)
5	+R6					2*	(3,R5)
6	+R7						3*

Trace back from R7: R7 <- R5 <- R2 <- R1.

Shortest path: R1 -> R2 -> R5 -> R7, total cost = 1 + 1 + 1 = 3. (The equal-cost alternative R1 -> R3 -> R6 -> R7 also has cost 3.)

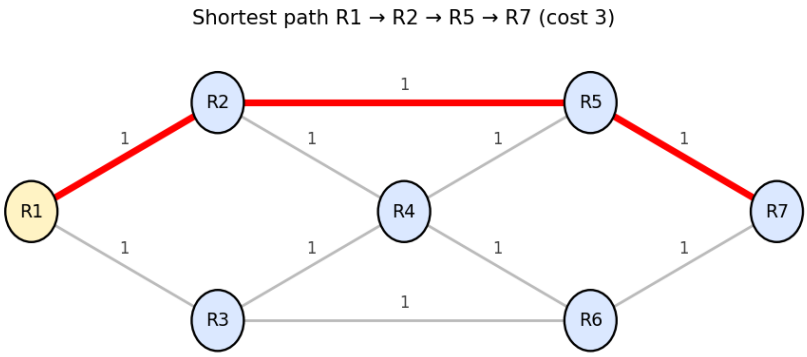


Figure 8: Shortest path from R1 to R7

b. Spanning tree and flooding (10 marks)

Spanning tree with R1 as root. A spanning tree of 7 nodes has 6 links and no loops. Taking the links through which each router first receives the packet (as in the figure): R1-R2, R1-R3, R2-R4, R2-R5, R3-R6, R5-R7.

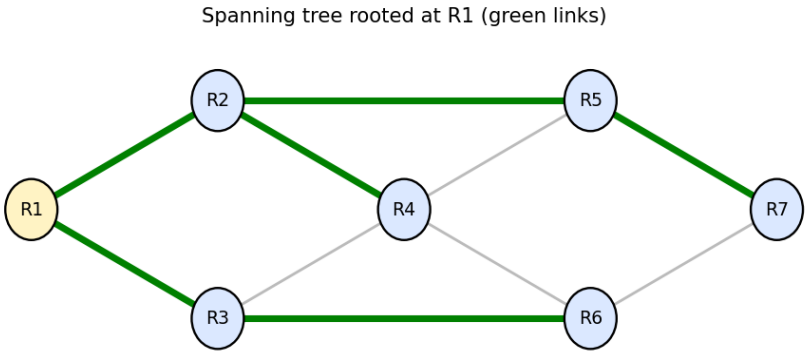


Figure 9: Spanning tree rooted at R1

Flooding without a spanning tree. Rule: a router forwards the first copy of a packet to all neighbours except those it received it from; duplicates are discarded (sequence number).

Time	Transmissions	Count
t = 0	R1->R2, R1->R3	2
t = 1	R2->R4, R2->R5, R3->R4, R3->R6	4
t = 2	R4->R5, R4->R6 (R4 heard from R2 and R3); R5->R4, R5->R7; R6->R4, R6->R7	6
t = 3	R7 received from R5 and R6, no other neighbour. R4, R5, R6 discard duplicates	0
Total		**12**

Of these 12 transmissions only 6 delivered the packet to a new router (R2, R3, R4, R5, R6, R7); the other 6 are duplicates.

Flooding along the spanning tree. Each router forwards only on tree links other than the one the packet arrived on:

Time	Transmissions	Count
t = 0	R1->R2, R1->R3	2
t = 1	R2->R4, R2->R5, R3->R6	3
t = 2	R5->R7 (R4, R6 are leaves)	1
Total		**6**

Comparison: flooding without the tree needs **12** transmissions versus **6** with the tree (= N-1, the minimum possible), a 50% saving. The tree eliminates duplicate copies and loops, and still reaches every router within the same 3 time steps. The cost is the need to compute and maintain the tree (and adapt when links fail).

Q9. Data Encryption Standard (DES) (5 marks)

DES is a symmetric-key block cipher (IBM, adopted by NIST as FIPS 46 in 1977). It encrypts a **64-bit plaintext block** into a 64-bit ciphertext block using a **56-bit effective key** (64 bits with 8 parity bits). The same key and algorithm are used for decryption, with subkeys applied in reverse order. It protects confidentiality of data in transit in network security.

Architecture

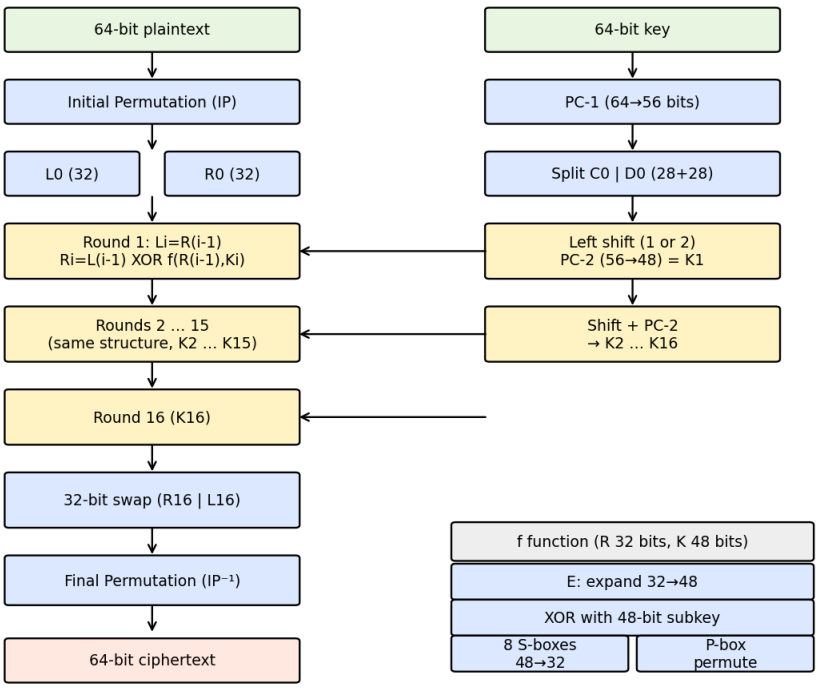


Figure 10: DES architecture

- **Initial Permutation (IP)** and **Final Permutation (IP^{-1})** - fixed bit permutations of the 64-bit block, inverses of each other.
- **16 rounds** of a Feistel structure; the block is divided into 32-bit halves L and R.
- **Key schedule:** PC-1 drops parity bits (64->56), splits into two 28-bit halves C and D, each round rotates them left by 1 or 2 positions and PC-2 selects a 48-bit round key K_i (16 subkeys).

Algorithm

- Apply IP to the plaintext block; split into L_0 and R_0 (32 bits each).
- For rounds i = 1 to 16: $\mathbf{L_i = R_{i-1}}$ and $\mathbf{R_i = L_{i-1}}$ **XOR** $\mathbf{f(R_{i-1}, K_i)}$.

- After round 16 swap the halves ($R_{16} \mid L_{16}$) and apply IP^{-1} to get the 64-bit ciphertext.

Round function f: (1) *Expansion* E: 32-bit R expanded to 48 bits by duplicating some bits; (2) *Key mixing*: XOR with 48-bit K_i; (3) *Substitution*: the 48 bits are split into eight 6-bit groups, each fed to an S-box giving 4 bits (outer two bits choose the row, inner four choose the column), producing 32 bits - this is the non-linear step; (4) *Permutation* P: the 32 bits are permuted.

Decryption uses the same circuit with the subkeys in the order $K_{16}, K_{15}, ..., K_1$.

Security remarks: the 56-bit key space (2^{56}) is now too small and can be brute-forced; DES has been replaced by **Triple DES** (three DES operations, 112/168-bit key) and then by AES. DES is still a good illustration of confusion (S-boxes) and diffusion (permutations) in network security.